

Deterministic Optimization

Linear Programming
Duality

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Weak Duality and Strong Duality

LP Duality

Learning Objectives for this lesson

- Discover weak duality
- Prove strong duality
- Identify their implications

LP Primal and Dual Pair

$$\begin{array}{ll} (P) & \min \quad \mathbf{c}^\top \mathbf{x} \\ & \text{s.t.} \quad \mathbf{Ax} = \mathbf{b} \\ & \quad \mathbf{x} \geq \mathbf{0}. \end{array}$$

$$\begin{array}{ll} (D) & \max \quad \mathbf{b}^\top \mathbf{y} \\ & \text{s.t.} \quad \mathbf{y}^\top \mathbf{A} \leq \mathbf{c}^\top \\ & \quad \mathbf{y} \text{ free.} \end{array}$$

Weak Duality

Given a primal LP in minimization, by the construction of the dual, the objective value of any feasible solution of the dual problem provides a lower bound to the primal objective cost

Theorem 1 (Linear programming weak duality). *If $\mathbf{x}$ is any feasible solution to the primal minimization LP, and $\mathbf{y}$ is any feasible solution to the dual maximization LP, then $\mathbf{c}^\top \mathbf{x} \geq \mathbf{b}^\top \mathbf{y}$.*

Implications of Weak Duality

Weak duality is a very useful property. With the weak duality theorem at hand, we can immediately get the following implications

1. If the optimal cost of the primal minimization problem is $-\infty$, then the dual maximization problem must be infeasible.
2. If the optimal cost of the dual maximization problem is $+\infty$, then the primal minimization problem must be infeasible.
3. Let $\mathbf{x}^*$ be feasible to the primal problem and $\mathbf{y}^*$ be feasible to the dual problem, and suppose $\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$, then $\mathbf{x}^*$ and $\mathbf{y}^*$ are optimal solutions to the primal and dual problems, respectively.

Strong Duality

- What does weak duality not imply?
 - Z_D may be *strictly* lower than Z_P
 - This indeed can happen for general optimization problem, but NOT for LP
- Strong duality:

Theorem 1 (Linear programming Strong Duality). *If a primal linear program has a finite optimal solution $\mathbf{x}^*$, then its dual linear program must also have a finite optimal solution $\mathbf{y}^*$, and the respective optimal objective values are equal, that is $\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$.*
- Strong duality is the single most important theorem in LP. Its proof is also very illuminating.

Prove Strong Duality

Proof. Assume the primal problem is a standard form LP (without loss of generality, we can transform any LP to standard form). Then the primal and dual problems are:

$$\begin{aligned} (P) \quad & \min \quad \mathbf{c}^\top \mathbf{x} \\ & \text{s.t.} \quad \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \quad \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

$$\begin{aligned} (D) \quad & \max \quad \mathbf{b}^\top \mathbf{y} \\ & \text{s.t.} \quad \mathbf{y}^\top \mathbf{A} \leq \mathbf{c}^\top \\ & \quad \mathbf{y} \text{ free.} \end{aligned}$$

If the primal problem has a finite optimal solution, then the simplex method with anti-cycling rule like the Bland's rule will terminate with an optimal solution $\mathbf{x}^*$ and the associated basis matrix $\mathbf{B}$. By the optimality of $\mathbf{x}^*$, we know the reduced costs are $\bar{\mathbf{c}}^\top = \mathbf{c}^\top - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A} \geq \mathbf{0}^\top$. Now let $(\mathbf{y}^*)^\top = \mathbf{c}_B^\top \mathbf{B}^{-1}$, then the reduced costs condition is the same as $(\mathbf{y}^*)^\top \mathbf{A} \leq \mathbf{c}^\top$, i.e. $\mathbf{y}^*$ is dual feasible. Also, $\mathbf{b}^\top \mathbf{y}^* = \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{b} = \mathbf{c}_B^\top \mathbf{x}_B = \mathbf{c}^\top \mathbf{x}^*$. By Weak Duality, $\mathbf{y}^*$ is an optimal solution to the dual problem, and the respective optimal objective values of primal and dual are equal. $\square$

Summary

- We learned:
 - Weak duality
 - Its implications
 - Strong duality